

Data Science for Energy System Modelling

Lecture 9: Sector-Coupling

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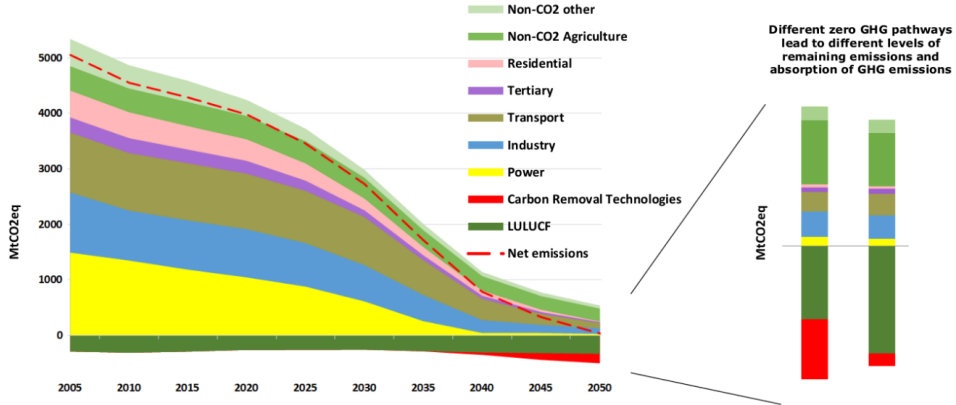
What is sector-coupling?

Why do we need it?

What impact will it have on electricity system operation?

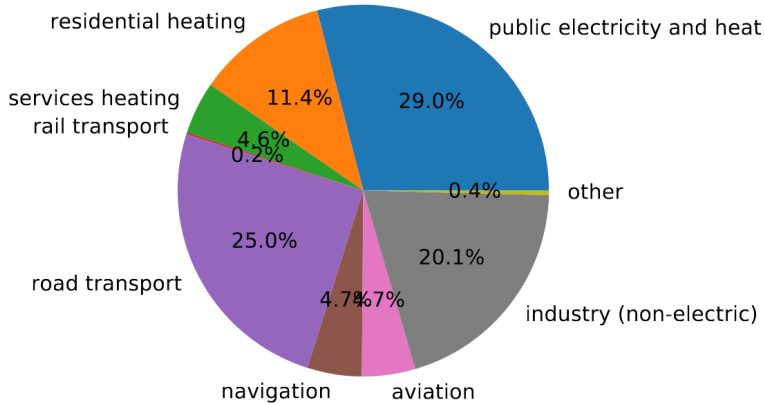
To reach net-zero emissions, all sectors need to decarbonise!

Paris-compliant 1.5° C scenarios from European Commission - **net-zero GHG in EU by 2050**

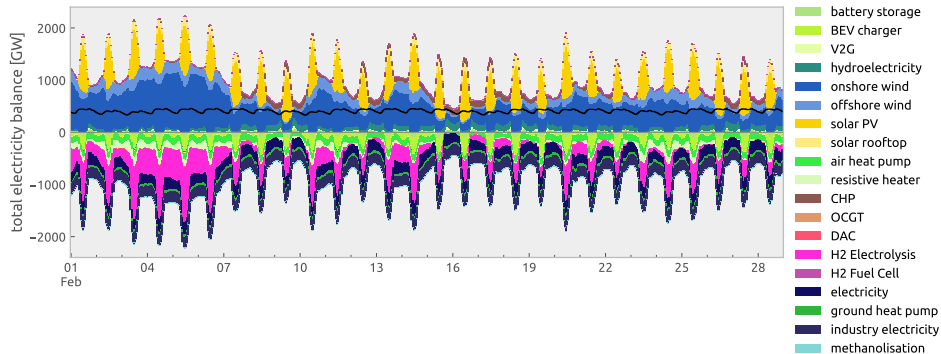


Public electricity/heat only responsible for a quarter of EU emissions.

EU28 CO₂ emissions in 2016 (total 3.5 Gt CO₂, 9.7% of global):

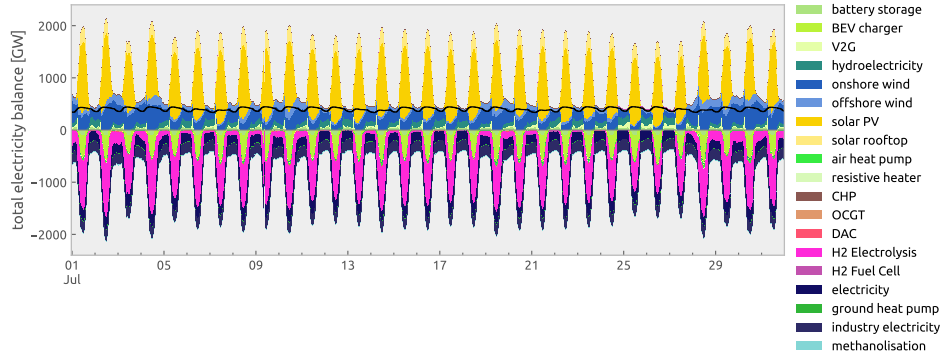


Electrification in many sectors affects operation of power system...



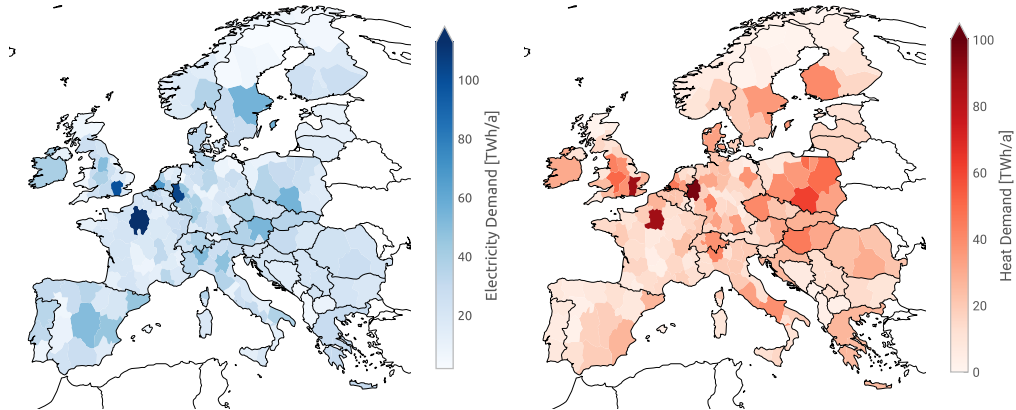
Sector-coupling adds a lot of electricity demand, requiring **much more wind and solar** buildout than if electricity sector alone were to be decarbonised.

...but also offers new opportunities for balancing renewables!

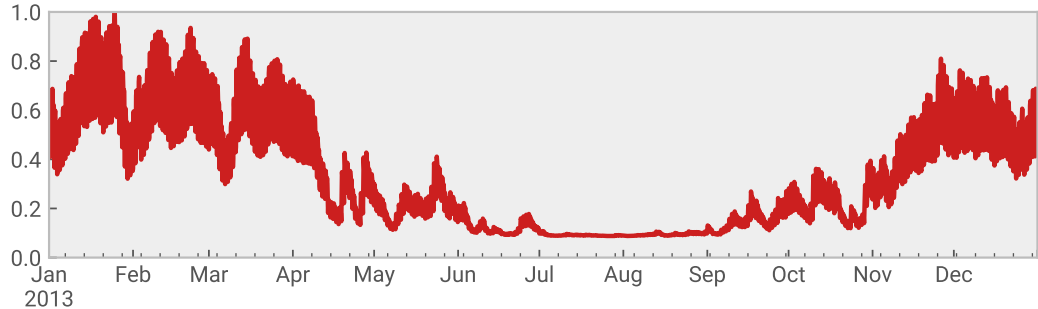


But sector-coupling also offers many ways to manage the variable renewable feed-in cost-effectively, through **demand-side management** (e.g. smart charging of electric vehicles) and **conversion to heat and chemical fuels** which are easier and cheaper to store

Like electricity, heating demand focused in densely populated areas.



Time series of heat demand show strong seasonal variation.



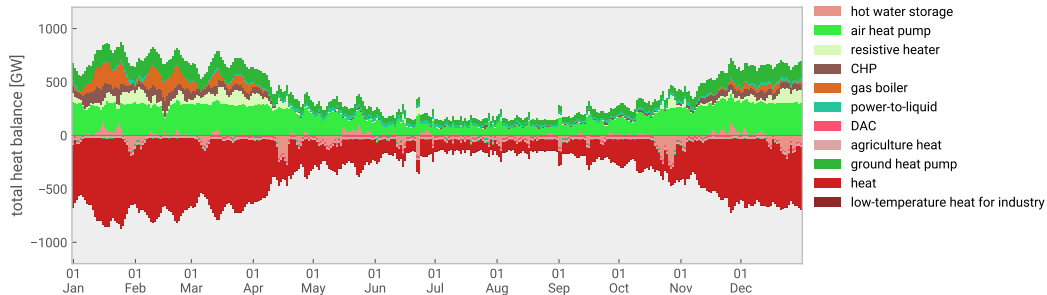
Compared to electricity demand, heating demand is strongly **seasonally peaked**, but also with weekly variations.

In summer, most heat demand is for **hot water** provision – in winter for **space heating**.

Technology stack for the heating sector

- heat pumps (air-sourced, ground-sourced)
- solar thermal (central, decentral)
- combined heat-and-power plants
- resistive heaters
- gas boilers (methane, hydrogen)
- pit thermal energy storage (central, long-term)
- hot water storage (decentral, short-term)
- waste heat from industrial processes
- combined with district heating networks
- building renovation (energy efficiency)

Example daily heat balance for net-zero emissions in Europe



There are difficult periods in winter with **low wind and solar generation**, **high space heating demand**, and **low air temperatures**, which are bad for air-sourced heat pumps!

In this case **gas boilers** and **CHP plants** jump in as backup. We also have some waste heat from e-fuel production in district heating.

How to calculate heating demand?

Heating degree day (HDD) quantifies the energy demand for heating a building. It is derived from the **outside air temperature** (T) and put in relation to a given **temperature threshold** (e.g. 15°C). The heating requirements are considered to be directly proportional to the number of HDD at that location.

$$\text{heat demand}(T) \propto \max\{0, 15^{\circ}\text{C} - T\}$$

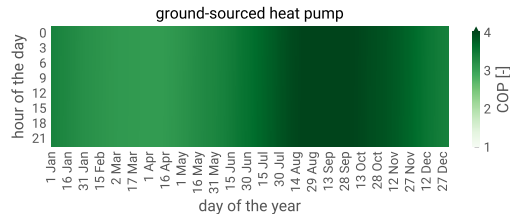
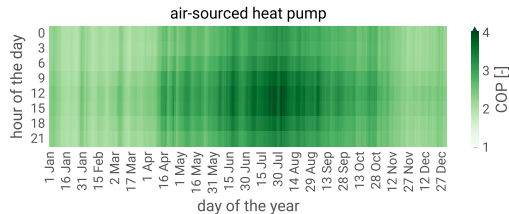
These daily profiles can be supplemented with typical hourly profiles for a weekday/weekend for downscaling.

Heat pumps

Heat pumps use electricity to move heat from ambient air or groundwater (source) into the homes (sink).

The **coefficient of performance (COP)** is temperature-dependent and defined as:

$$COP = \frac{\text{thermal energy moved from source to sink}}{\text{input work (electricity)}} \propto \frac{1}{T_{\text{sink}} - T_{\text{source}}}$$



Heat pump performance is determined from measurements.

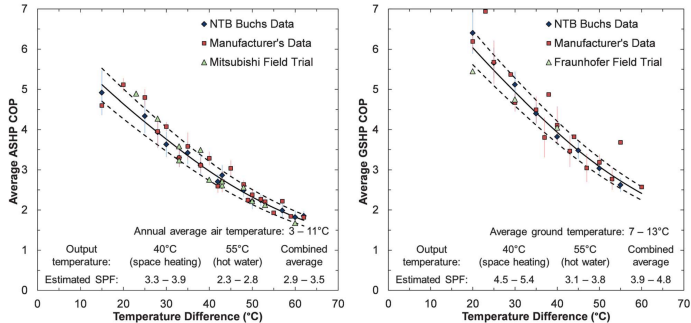


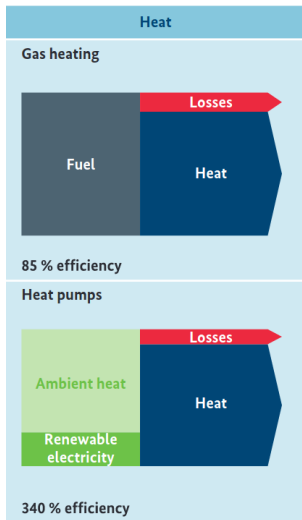
Fig. 9 Average heating coefficient of performance for air and ground source heat pumps (left and right, respectively) based on data taken from industrial surveys and field trials.^{31,40–42} The inset tables show the expected performance for UK conditions.

For instance, for an air-sourced heat pump an approximated function could look like this

$$COP(\Delta T) = 6.81 + 0.121\Delta T + 0.000630\Delta T^2$$

where ΔT is the difference between the air (source) and water storage (sink) temperature.

Why heat pumps work in favour of decarbonisation?

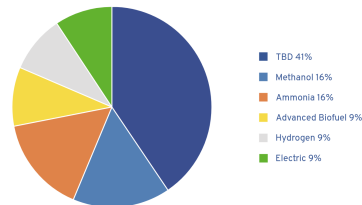


Technology stack for the transport sector

Land transport

- battery electric vehicles
- fuel cell electric vehicles (hydrogen)
- catenary trucks (=like trains)
- internal combustion engines
- smart charging
- vehicle-to-grid

Shipping



Aviation

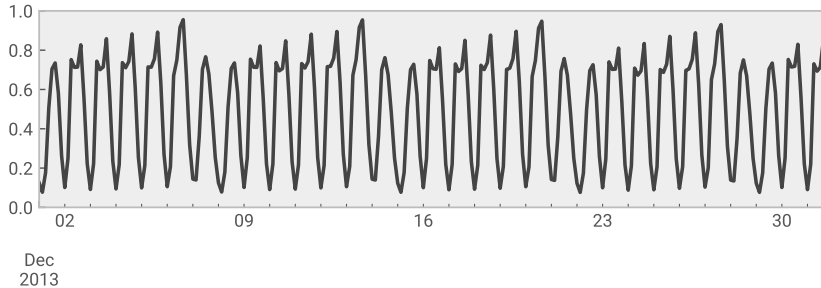
- electricity (short-haul)
- hydrogen (medium-haul)
- synthetic kerosene (long-haul)

Source:

<https://www.globalmaritimeforum.org/content/2022/11/The-2022-Annual-Progress-Report-on-Green-Shipping-Corridors.pdf>

To model demand-side management with electric vehicles,...

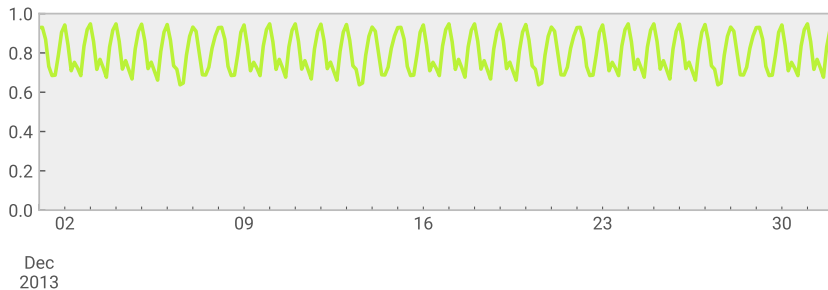
...need to know the typical mobility behaviour of people ...



(this profile is strongly peaked)

To model demand-side management with electric vehicles,...

...need to know the typical availability of electric vehicles for flexible operation ...

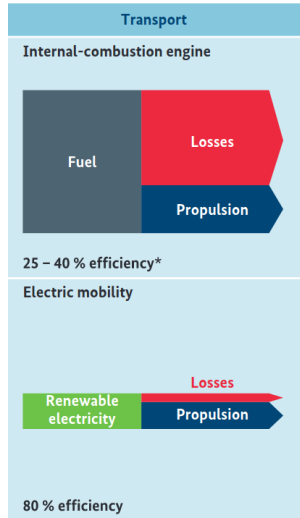


(this profile shows normalised availability for charging at home)

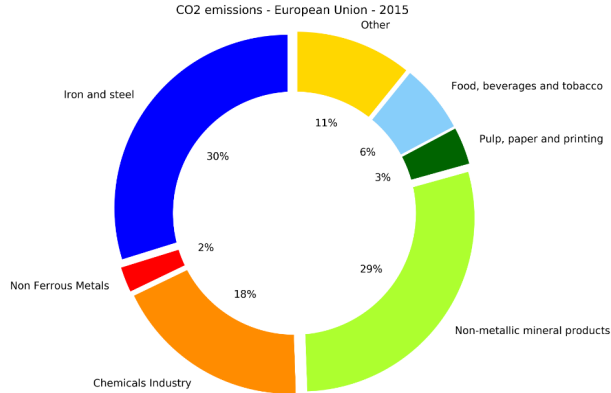
(need also technical parameters like charging speed and battery capacity)

(need also social parameters like when battery should be fully charged)

What is argument for electric vehicles from an energy systems view?



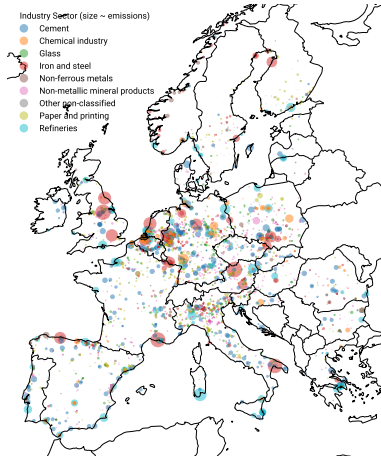
Which industry sector is responsible for most emissions?



- Non-ferrous metals: mainly aluminium, but also copper, lead etc.
- Non-metallic minerals: mainly cement, ceramics and glass
- Emissions come from combustion of fossil fuels for heat, as well as **process emissions** from chemical reactions

Industry: not only energy but also non-energy demands (feedstocks) are relevant.

Spatial distribution of different industry sectors in Europe.

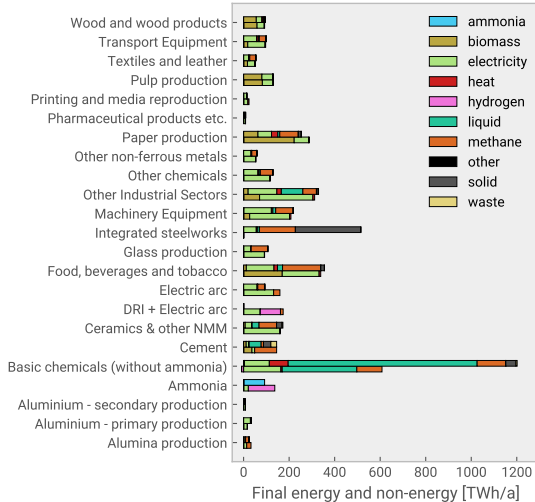


- **Agglomeration of industries** in certain regions (e.g. Ruhr valley, Rotterdam) can create challenges for future energy supply depending on level of network reinforcements
- **Industry relocation**, where certain energy-intensive industries migrate to regions with cheap renewable production, is also hotly discussed at the moment
- Because of **process emissions** which could be captured, industrial sites are also interesting for co-location with synthetic fuel production processes which require CO₂ as inputs.

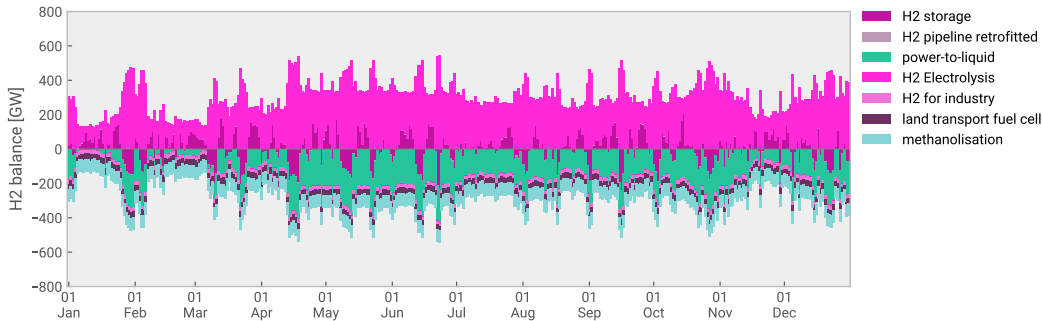
Technology stack for the industry sector

- carbon capture of process emissions
- circular economy practices (e.g. recycling of plastics and steel)
- demand reductions
- industry relocation within Europe / outside of Europe
- process switching (e.g. in steelmaking from coke furnaces to electric arc furnaces and hydrogen-based DRI)
- fuel switching (e.g. synthetic/biogenic fuels rather than fossil fuels)
- imports of renewable fuels and feedstocks

Exemplary composition of energy carrier demands by industry sector.



H₂ – intermediate product between electricity and synfuels.

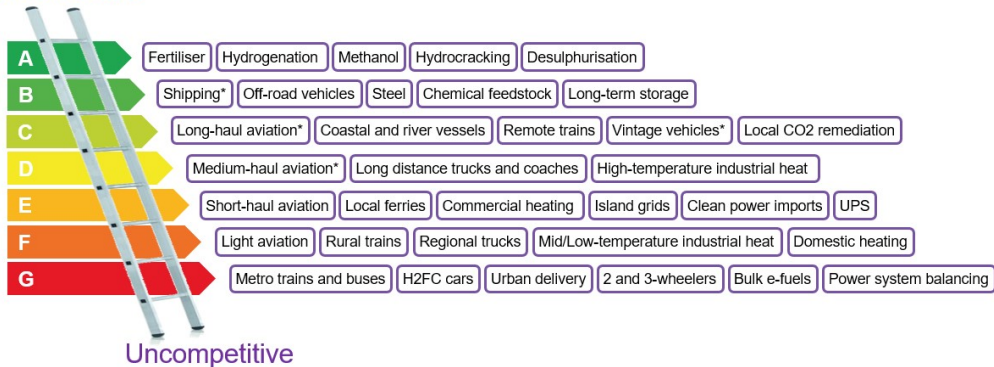


Relatively **few end-uses** of hydrogen (land transport, industry).

Most of the hydrogen is produced to produce **hydrogen derivatives**, e.g. high-value chemicals, kerosene, methanol, methane etc.

The hydrogen ladder – unavoidable and uncompetitive uses

Unavoidable

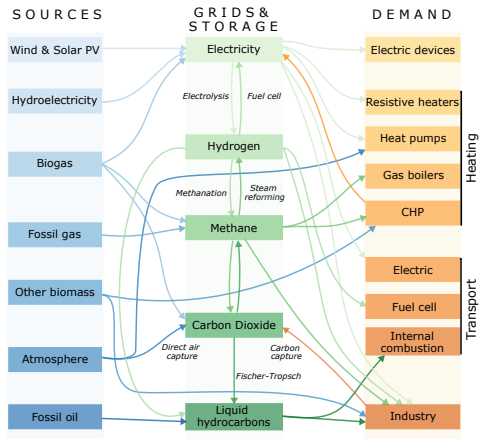


* Via ammonia or e-fuel rather than H2 gas or liquid

Source: Liebreich Associates (concept credit: Adrian Hiel/Energy Cities)

Many interactions between sectors that need to be modelled...

... each with sector-specific **energy demand patterns** and **flexibility options**.



- Electricity sector
- Transport sector
- Heating sector
- Industry sector
- International shipping and aviation
- Industrial feedstocks and biomass
- Electricity, hydrogen and gas networks
- Carbon management strategy!

Up next

Next Thursday, 1: Sector-coupling with pypsa.

Next Thursday, 2: Consultation hour for group assignment.